Design Doc: Giddly Offline-First Architecture

To: Giddly Talent Team From: Akhila Thammenenwar Date: November 16, 2025 Subject: Feature Proposal - Smart Offline Mode

1. The Problem

Based on my review, the Giddly app is highly dependent on a stable internet connection. During testing, I encountered "Network error" and "Something went wrong" screens when the network was weak.

This is a critical issue for a social event app, as users are frequently in low-connectivity areas (like stadiums, conference centers, or subways) when they need the app most.

2. The Solution: "Offline-First" Architecture

My proposal is not a new feature, but a critical enhancement to the app's reliability. By implementing an "offline-first" architecture, we can provide a seamless user experience, even when the network fails.

My solution has three parts:

Proactive Caching: When the app has a good connection, it proactively saves essential data (like event details, user contacts, and calendar info) to a local database on the user's device.

Graceful Handling: When offline, the app loads data from the local cache instead of showing an error. It displays a simple, non-intrusive banner (like "You are offline. Showing saved data.")

Action Queuing: If a user performs an action (like responding to an event) while offline, the app saves this action and automatically syncs it with the server once the connection returns.

3. Technical Design (How to Build It)

Component	Technology / Approach
Local Database	**Hive** (or sqflite). Hive is a lightweight, high-performance NoSQL database for Dart/Flutter. It's perfect for caching the JSON responses from the Giddly API.
Network Detection	**connectivity_plus (Flutter package).** This package provides a stream that notifies the app *instantly* when the network state changes (online to offline, or vice-versa), allowing the UI to react.
API/State Logic	**State Management (like Provider or Riverpod).** The app's logic would be wrapped in a way that it first checks the network. If online, it fetches from the API and saves to Hive. If offline, it *only* reads from Hive.

4. Prototype

To prove this concept, I have built a functional prototype. It's a simple Flutter app that:

1.Fetches mock data from an API.

2.Saves that data to a local Hive database.

3.When the app is restarted in offline mode (or the refresh button is pressed), it loads the data from the cache and displays an "offline" banner.

GitHub Repository Link: https://github.com/Akhila21-6/giddly-offline-prototype